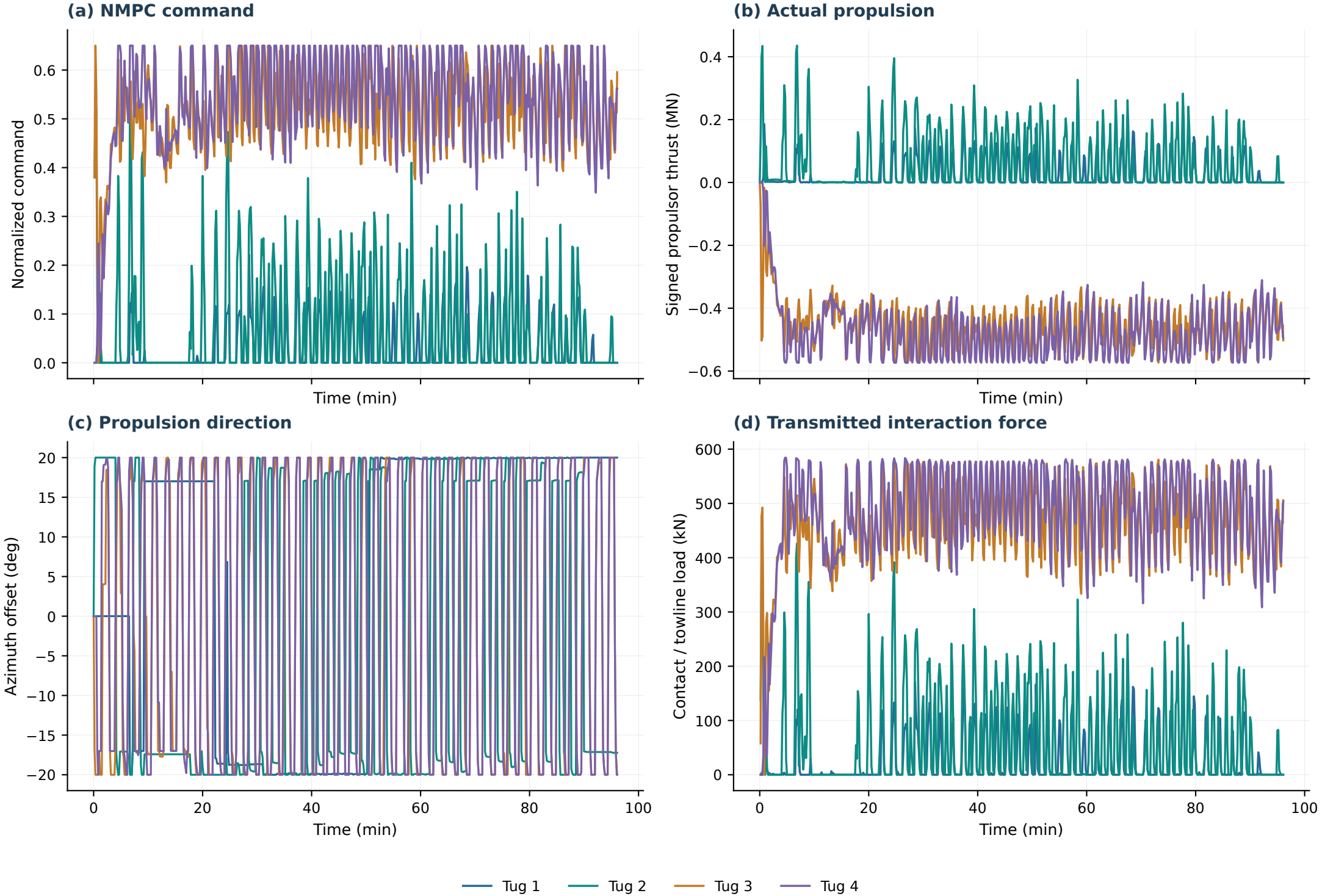

(a) NMPC command
Normalized command
Time (min)
(b) Actual propulsion
Signed propulsor thrust (MN)
Time (min)
(c) Propulsion direction
Azimuth offset (deg)
Time (min)
(d) Transmitted interaction force
Contact / towline load (kN)
Time (min)
Tug 1
Tug 2
Tug 3
Tug 4